Solutions

Question 1.1:

(a) $4^2 = 16$

(b) $5^3 = 125$

(c) $\frac{2}{5}^3 = \frac{8}{125}$

Question 1.2:

(a) $a * a * a * a = a^4$

(b) $(yz) * (yz) * (yz) = (yz)^3$

(c) $\left(\frac{a}{f}\right) * \left(\frac{a}{f}\right) = \left(\frac{a}{f}\right)^2$

Question 1.3:

(a) $5^{13} * 5^3 = 5^{16}$

(b) $8^{13}/8^5 = 8^8$

(c) $\left(a^3\right)^4 = a^{12}$

(d) $\frac{y^7}{y^3} = y^4$

Question 1.4:

(a) $\frac{\exp(-8)}{\exp(-10)} = \exp(2)$

(b) $\frac{\exp(3)}{\exp(-15)} =$

(c) $\frac{\exp(-1)\exp(-1)}{\exp(-5)} = \exp(3)$

(d) $\frac{\exp(-3)^2}{\exp(-6)} = 1$

Question 1.5:

(a) $\log_{10}\left(4b^2\right) = 3 \implies b = 15.811.$

(b) $\exp\left(\frac{2b-1}{3}\right) = 5 \implies b = 2.914$

Question 1.6:

(a) $\exp(2t) = 5 \implies t = 0.805$

(b) $\exp(5t) \times \exp(-t) = 5 \implies t = 0.402$

(c) $\exp(10t) \div \exp(-t) = 5 \implies t = 0.146$

(d) Two equations: $\exp(4t) = 0.1N$ and $\exp(t) = N \implies t = -0.768$

Question 1.7:

(a) $x = \log_3(3) \implies x = 1.000$

(b) $x = \log_5(10) \implies x = 1.431$

(c) $x = \log_2(8) \implies x = 3.000$

(d) $x = \log_3(59049) \implies x = 10.000$

Question 1.8:

(a) $y = 2x + 4x + 5x \implies x = \frac{y}{11}$

(b) $y = \frac{1}{2x+4x+5x} \implies x = \frac{1}{11y}$

(c) $y = \frac{1}{2x^2+4x^2+5x^2} \implies x = \pm\sqrt{\frac{1}{11y}}$

(d) $y = \frac{x}{2x^2+4x^2+5x^2} \implies x = \frac{1}{11y}$

(e) $y = \sqrt{x+1} \implies x = y^2 - 1$

(f) $M = \frac{\pi\rho x^3}{6} \implies x = \sqrt[3]{\left(\frac{6M}{\pi\rho}\right)}$

Question 1.9: Expand the following:

(a) $3\left(x^2 + 3y\right) \implies 3x^2 + 9y$

(b) $7x\left(-\frac{1}{x} + y^3\right) \implies -7 + 7xy^3$

(c) $3xy\left(x + \frac{2}{xy}\right) \implies 3x^2y + 6$

Question 1.10: The equation describing radioactive decay is:

$$\frac{dN}{dt} = -\lambda N$$

where N is the number of unstable nuclei, t is time and λ is called the decay constant. This is a differential equation. The solution to this differential equation is given by the following:

$$N(t) = N_0 e^{-\lambda t}$$

Here we can simply set the value of N to be half of N_o. Then, by using common laws regarding re-arranging logarithms, we can use the following procedure:

$$\begin{aligned} \frac{N_o}{2} &= N_o e^{-\lambda t_{1/2}} \\ ln\left(\frac{1}{2}\right) &= -\lambda t_{1/2} \\ ln\left(\left(\frac{1}{2}\right)^{-1}\right) &= \lambda t_{1/2} \\ \lambda &= \frac{ln2}{t_{1/2}} \end{aligned}$$

where $t_{1/2}$ is the half-life of the sample.

Question 1.11:

(a) 68.2°

(b) 6.53

(c) 26.13

(d) 0.20 radians

Question 1.12:

1. $F1 = mg\cos(\theta)$
2. $F2 = mg\sin(\theta)$

Question 1.13:

(a) - (b) Filled in table:

x	y	Backward	Forward	Central	Analytical
2	8.53	–	–	–	–
4	6.00	-1.26	-0.89	-1.07	-1.08
6	4.23	-0.89	-0.48	-0.68	-0.69
8	3.27	-0.48	-0.06	-0.27	-0.27
10	3.16	-0.06	0.37	0.16	0.16
12	3.91	0.37	0.81	0.59	0.59
14	5.52	0.81	1.24	1.02	1.02
16	8.00	1.24	1.68	1.46	1.46
18	11.35	1.68	2.12	1.90	1.90
20	15.58	–	–	–	–

(c) Looking at the raw differences:

x	Backward	Forward	Central
4	-0.18	0.20	0.01
6	-0.20	0.21	0.00
8	-0.21	0.21	0.00
10	-0.21	0.22	0.00
12	-0.22	0.22	0.00
14	-0.22	0.22	0.00
16	-0.22	0.22	0.00
18	-0.22	0.22	0.00

and also looking at the percentage difference:

x	Backward	Forward	Central
4	16.55	18.18	0.81
6	29.44	30.31	0.44
8	78.28	79.34	0.53
10	134.97	135.99	0.51
12	36.63	36.80	0.09
14	21.23	21.30	0.03
16	14.95	14.99	0.02
18	11.54	11.56	0.01

We can see that the central differences method is very accurate. Recall from lectures that the central differences method is more accurate for smooth functions, though this accuracy can vary depending on the shape of our curves.

Question 1.14:

(a) Filled in table:

t	$N(t)$	**Numerical** (ΔN)	**Analytical** ($-\lambda N(t)$)
0	1000.00	-22.84	-23.10
1	977.16	-22.31	-22.57
2	954.85	-21.80	-22.06
3	933.05	-21.31	-21.55
4	911.74	-20.82	-21.06
5	890.92	-20.34	-20.58
6	870.58	-19.88	-20.11
7	850.70	-19.43	-19.65
8	831.27	-18.98	-19.20
9	812.29	-18.55	-18.76
10	793.74	-18.13	-18.34

(b) The agreement is pretty good in this case. Note that when using the approximate method the step size chosen must be small enough. You will only figure this out by trial and error (e.g. try a step size and then try a smaller step size to see if the results are the same).

Question 1.15:

(a) - (b) The table below gives the solutions to both parts where the time difference moves from 1 to 2 months. Notice how just using these derived gradients could be misleading. We need to see longer term trends to infer any global changes.

year	month	average (ppm)	forward	central	forward	central
2021	4	419.05	–	–	–	–
	5	419.13	–	–	–	–
	6	418.94	-1.980	-1.085	-2.235	-1.145
	7	416.96	–	–	–	–
	8	414.47	–	–	–	–
2022	4	420.23	–	–	–	–
	5	420.99	–	–	–	–
	6	420.99	-2.090	-1.045	-1.900	-0.760
	7	418.90	–	–	–	–
	8	417.19	–	–	–	–

Question 1.16:

(a) The table below gives the solutions for the numerical approximation of the derivative of $f(x)$ for $x = 1$ using Euler's Backward, Euler's Forward, and the Central Differences Methods using a step size of $\Delta x = 1$, $\Delta x = 0.1$, $\Delta x = 0.01$, and $\Delta x = 0.001$. We observe that $f(x) = 14$ when $x = 1$.

Δx	$f(x-\Delta x)$	$f(x+\Delta x)$	**Backward**	**Forward**	**Central**
1	16.000	22.000	-2.000	8.000	3.000
0.1	13.849	14.251	1.510	2.510	2.010
0.01	13.980	14.021	1.950	2.050	2.000
0.001	13.998	14.002	1.995	2.005	2.000

(b) We observed that $f'(x) = 2$ when $x = 1$. The table below gives differences between the numerical and analytical derivatives. We can see that as we increase the step size that the accuracy of the numerical derivative increases also. This is one of the considerations when finding derivative numerically, increasing the step size increases the accuracy but also increase the computation, so there is a trade off in big applications.

Δx	**Backward**	**Forward**	**Central**
1	-4.000	6.000	1.000
0.1	-0.490	0.510	0.010
0.01	-0.050	0.050	0.000
0.001	-0.005	0.005	0.000